PLUSQUAM MODEL COMPARISON

Model Comparison Summary

Qwen and Groq on the same 30 assessments: quality, speed and cost.

30	10	3	4
assessments per model	synthetic project cases	runs per case	measures compared

APPLICATION	Plusquam assessment platform
MODELS	Qwen qwen-plus · Groq GPT-OSS-120B
REPORT DATE	26 October 2026

RESULTS

Qwen and Groq, side by side

Same prompt, same inputs, three runs per case.

	QWEN qwen-plus	GROQ GPT-OSS-120B
QUALITY		
Valid assessment structure	● 30 of 30	21 of 30
Same verdict across repeats	● 9 of 10 cases	6 of 10 cases
Unsupported paper references	● 0 outputs	12 outputs
SPEED		
Median response time	21.3 s	● 1.9 s
Slowest 5% (p95)	24.1 s	● 2.8 s
COST		
Estimated cost, 30 runs	$0.064	● $0.019
Estimated cost per assessment	about $0.002	● about $0.0006

SUMMARY

Qwen returned a valid, consistent result in every run. Groq answered faster and at lower cost, but 9 of 30 of its outputs failed the structure check.

TEST SETUP

Ten synthetic project cases, three runs each, with the same prompt, temperature 0.3 and JSON output. Groq GPT-OSS-120B was used because the previous Llama 3.3 70B model was no longer accessible. Costs are estimates from token usage and list prices.